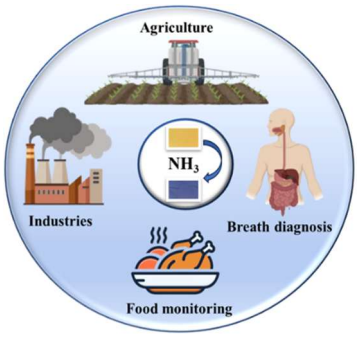
Dye-Integrated Electrospun Nanofiber Sensors for Visual Detection of Ammonia: A Portable and Low-Cost Approach Using PCL-BPB/BCG Composites

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***Abstract*- Ammonia (NH_3) is widely used in agriculture, domestic, and industrial sectors. Its extraordinary volatility and toxicity can cause significant environmental and health concerns. We report a simple and inexpensive colorimetric nanofiber sensor by electrospinning polycaprolactone (PCL) with Bromocresol Green (BCG) and Bromophenol Blue (BPB) dyes for ammonia detection. The nanofibers were examined using field emission scanning electron microscopy (FE-SEM) and rheological analysis. SEM confirmed the nanofibers' smooth and uniform shape, while rheological studies revealed that adding dyes of different concentrations significantly reduced the viscosity of the spinning solution due to plasticization effects. The proposed sensor demonstrated clear colorimetric reactions to ammonia gas, presenting its potential for portable, on-site environmental and occupational safety detection applications.**



Keywords*—*Ammonia gas, Colorimetric Sensor, Electrospinning, Nanofibers

I. Introduction

Ammonia (NH_3) is widely used as a chemical in food processing, agriculture, textiles, and commercial industries [1],[2]. It is the primary precursor for synthesizing amino acids and nucleotides, a prominent nitrogen source. The deposition of ammonia is increasing on land and in aquatic systems through bacterial processes and ruminant digestion in natural environments. It is also utilized and released in agricultural fields, industries, and domestic activities. Due to this reason, ammonia production globally was estimated to be 235 million metric tons in 2019 [3] and is expected to be 290 million metric tons by 2030 [4]. The emission of this ammonia from different fields extensively affects human health and lifespan expectancy [3], and its high volatility and toxicity cause significant health and environmental risks at lower concentrations [5],[6]. The 50-100 ppm level of ammonia gas irritates the nose, eyes, and throat of humans and animals [7], while exposure to 300 ppm is considered an immediate danger to human health [8]. Trace amount of ammonia gas is produced from the decomposition of ammonium nitrate (NH_4NO_3), which is widely present in multiple fertilizers. In 2020, a massive explosion occurred in Beirut, Lebanon, due to the association of NH_4NO_3 from fertilizers, which caused major human deaths [9]. Spotting low concentrations of NH_3 is critical, mainly for prior information in environmental and occupational safety. So many conventional techniques are available for detecting NH_3, but they lack sensitivity and require complex instrumentation. A simple, sensitive, and affordable system for monitoring the NH_3 volatile gas and its derivatives is required. Colorimetric sensor arrays (CSAs) employ a variety of chemically responsive dyes to provide a pattern-based response to target gases, improving selectivity and sensitivity[10], [11]. This study aims to develop a cost-effective and straightforward method to fabricate a colorimetric sensor with an easy visual readout and integration into a portable platform.

II. Experimental Details

The colorimetric sensor array for detecting and quantifying NH_3 is fabricated using electrospinning technology. The process is as follows: (i) A combination mixture consists of 10% Poly Caprolactone (PCL) polymer, chloroform, and methanol (3:1), (ii) Bromocresol green (BCG) and bromophenol blue (BPB) dyes are added to the polymer solution in different concentrations, (iii) The prepared polymer and dye solution is transferred to a disposable plastic syringe fitted with a 24-gauge metal needle, (iv) Nanofibers were collected on an aluminum sheet by applying an 18 KV voltage with a constant flow rate of 2 mL/hr to realize the sensors. Materials were procured from Sigma-Aldrich PCL (80 KDa),

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SRL Private Limited BCG dye, and Thermo Fisher Scientific BPB dye.

A. Sensor surface characterization

Field Emission Scanning Electron Microscopy (FE-SEM) images were collected to know about the surface and fiber morphology of the electrospun nanofiber samples using an Ultra55 FE-SEM Karl Zeiss mono instrument, and samples were sputter-coated with a 10 nm layer of gold for 1 minute using a JEOL sputtering tool before SEM imaging to enhance the surface conductivity.

B. Viscosity

The spinning solution's viscosity is measured using a Modular Compact Rheometer – 302 (MCR–302) tool under controlled shear conditions. A cone plate with a 40 mm diameter, gap 0.08 mm, is used in contact with a flat plate, ensuring uniform shear across the sample. Measurements were performed at 25 °C across a shear rate of 1 to 1000 s^{-1}.

C. Hardware system setup

We have developed a HepaHue system (A platform integrated with a colorimetric sensor array for VOC detection) and integrated it with the CALC Web app to validate the performance of the colorimetric sensor array. The color change of the sensor array has been monitored and recorded continuously after exposure to ammonia gas of different concentrations.

The whole system is divided into the following two parts:

1) HepaHue

The HepaHue block diagram is given in Fig. 1. The system is powered by an AC to DC adapter that converts 220V AC to 12V DC, which is further regulated to 5.1V DC using a DC-DC converter to power the Raspberry Pi Zero W2 and peripheral components. The Raspberry Pi is the main controller, interfacing with various modules via General-Purpose Input/Output (GPIOs) and Flat Flexible Cable (FFC) connections. It runs embedded software, including image capturing, image processing, Graphical User Interface (GUI), vacuum control, and data logging. The system includes a vacuum pump, controlled through a motor driver circuit, to draw gases out of the chamber. White Light Emitting Diodes (LED), managed by an LED driver circuit, illuminate the disposable colorimetric patches, and the resulting color changes are captured using a Raspberry Pi Camera Module. A BME680 sensor monitors environmental parameters, and a GUI is displayed on a 4.3-inch High-Definition Multimedia Interface (HDMI) touchscreen. Processed data can also be uploaded to cloud storage for further analysis or record keeping. Primary and Secondary Printed Circuit Boards (PCBs) are custom-made. In addition to integrating embedded systems, image acquisition is carried out at multiple stages: confirming proper patch array placement, during the pre-exposure analysis, and post-exposure analysis after VOCs have been introduced into the chamber, as shown in Fig. 3.

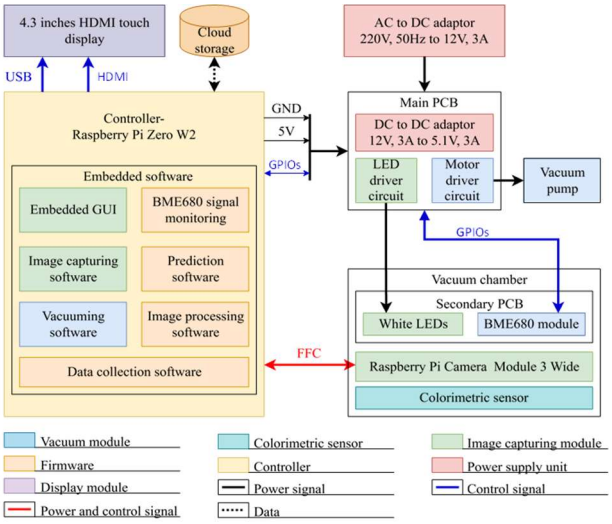


Fig. 1. Block diagram Schematics of HepaHue.

2) CALCWebApp

CALC stands for Colorimetric Analysis of Liver Cirrhosis. This is a web application that manages and stores data that is collected by HepaHue. This application uses the MERN (MongoDB, ExpressJS, ReactJS, NodeJS) stack, whose block diagram can be seen in Fig. 2.

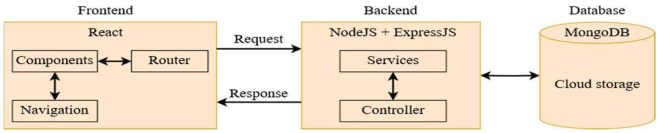


Fig. 2. Block diagram of CALC Web App.

The average red, green, and blue (RGB) components are extracted periodically from the captured images for each colorimetric patch. Let $R_i(t)$, $G_i(t)$, and $B_i(t)$ represent the average red, green, and blue intensity values, respectively, for the i-th patch at time t. The initial RGB values at time t = 0 are denoted as $R_i(0)$, $G_i(0)$, and $B_i(0)$. To quantify the color change of each patch over time, the Euclidean distance in the RGB color space is computed between the current RGB values and their corresponding initial values. This distance measures the magnitude of color change and is calculated.

$$Di(t) = \sqrt{\{[R_i(t)-R_i(0)]^2 + [G_i(t)-G_i(0)]^2 + [B_i(t)-B_i(0)]^2\}}$$

where:

- $D_i(t)$ is Euclidean distance for the i-th patch at time t.
- $R_i(t)$, $G_i(t)$, $B_i(t)$ are the mean RGB values at time t.
- $R_i(0)$, $G_i(0)$, $B_i(0)$ are the mean RGB values at the initial time t =0.

This distance $D_i(t)$ serves as a robust metric to monitor and analyze the temporal evolution of color change due to chemical interaction with the breath sample.

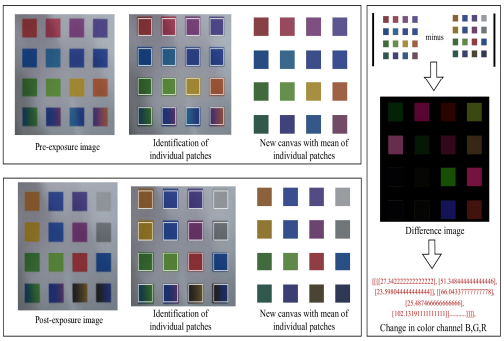


Fig. 3. Steps involved in image acquisition and processing.

III. RESULTS AND DISCUSSIONS

After developing a colorimetric nanofiber sensor, material characterization (SEM) was performed to study the morphology of Electrospun nanofibers, which were prepared using different ratios of PCL polymer and dyes, respectively. Figure 1 shows the morphology of nanofibers. SEM images confirmed that the nanofibers are smooth and continuous with an average diameter of 217.595 nm.

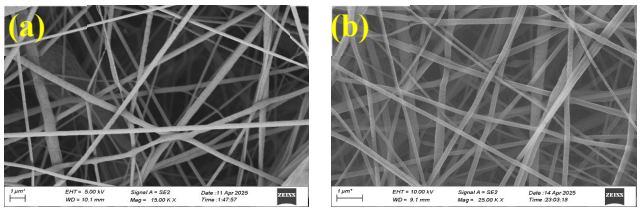


Fig. 4. (a) SEM image of a bare nanofiber; (b) SEM images of BCG dye-treated nanofibers.

After adding dyes (BCG & BPB) to the PCL polymer, a change in viscosity was observed and recorded by the rheometer as shown in Figure 2. The change in viscosity is due to the dyes acting as plasticizers, increasing the free volume of the polymer by decreasing crystallinity and chain mobility. Resulting in lower internal resistance and polymer thinning behaviour at higher concentrations.

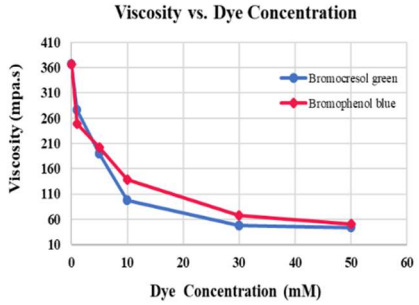


Fig. 5. Viscosity profile of nanofibers with and without the addition of BCG and BPB dyes, showing reduced viscosity with increase in dyes concentration.

The test was conducted primarily to evaluate the performance of Ammonia indicator dye patches made from nanofibers embedded with bromocresol green and bromophenol blue. These patches were prepared using different dye concentrations (1 mM, 5 mM, 10 mM, 30 mM, and 50mM).

To understand the selectivity of the sensor array towards a mixture of VOCs, a mixture consisting of 2 µL of Ammonia, Ethanol, Methanol, and Acetone solution, each vaporized separately in a 2-liter Tedlar bag filled with nitrogen gas, was used. The array was exposed for 2 minutes with images captured every 10 seconds at room temperature.

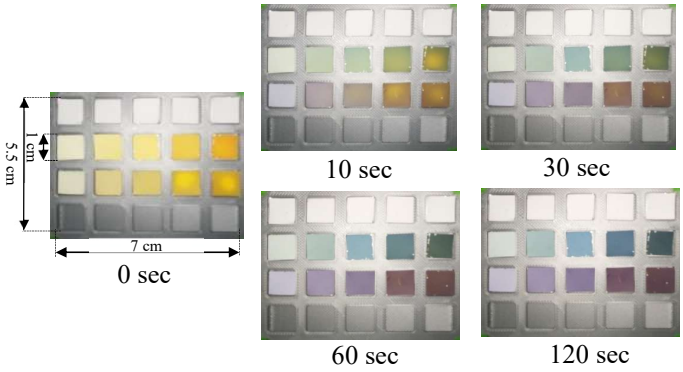


Fig. 6. Color change in the colorimetric sensor array upon exposure to ammonia gas for 0 -120 seconds.

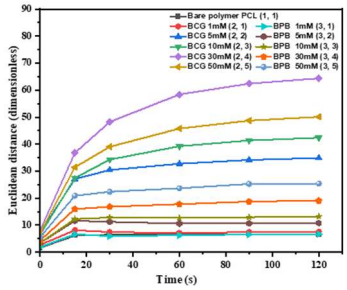


Fig.7.Colorimetric sensor array response to ammonia gas exposure over time.

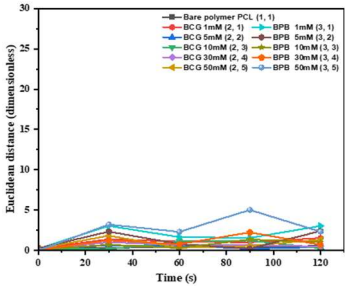


Fig.8. Colorimetric sensor array response to acetone gas exposure over time.

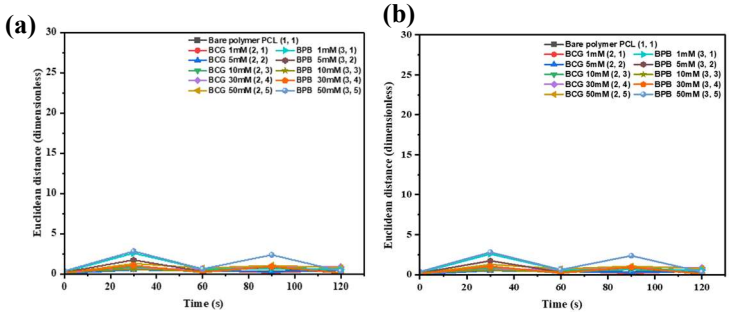


Fig. 9. Colorimetric sensor array response to gas exposure over time. (a) Ethanol, (b) Methanol.

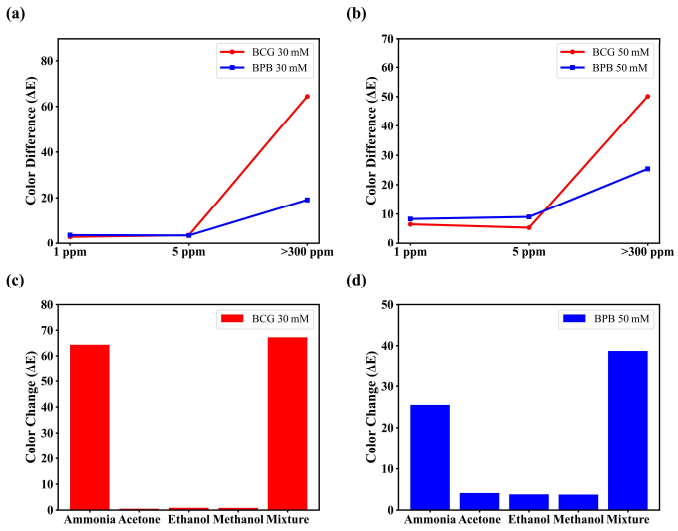


Fig. 10. Sensor array response to different ammonia gas concentrations of BCG and BPB dyes: (a) 30 mM, (b) 50 mM. Selectivity plots for dyes (c) BCG (30 mM) and (d) BPB (50 mM) toward ammonia, acetone, ethanol, methanol, and a mixture of vapors.

Table 1 shows the images of a patch array exclusively selective to ammonia.

TABLE 1. Pre and post-exposure images of the colorimetric patch array in the presence of Ammonia, Acetone, Ethanol, and Methanol vapors.

Biomarker	Pre-Exposure Image	Post-Exposure image
Ammonia		
Acetone		
Ethanol		
Methanol		

The following observations are made from the plots (Fig. 6-10). Bare polymer does not change color when reacting with ammonia gas. Bromophenol blue has a faster response time than bromocresol green, but the Euclidean distance of bromocresol green is higher. No significant color change was observed for acetone, ethanol, or methanol vapors.

IV. CONCLUSIONS

This study presents a low-cost, disposable, colorimetric sensor array patch fabricated using the electrospinning technique and integrated into a portable Hepa Hue system and CALCWeb App for real-time detection and quantification of ammonia gas. The sensors showed a wide range of ammonia concentrations (1 ppm to beyond 200 ppm) detection capabilities with a limit of detection (LOD) of 1 ppm in 5 minutes. We envisage enhancing the sensitivity of the sensor array, making it feasible for healthcare applications using a non-invasive approach.

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